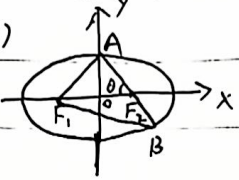
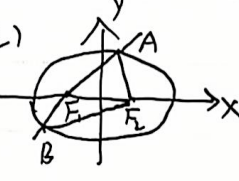
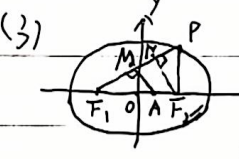


离心率问题

1. (1)  由 $\overrightarrow{AF_2} = 3\overrightarrow{F_2B}$ 则设 $|BF_2| = m (m > 0)$, 则 $|AF_2| = 3m$
 则由第一定义: $|AF_1| + |AF_2| = 2a$, $|BF_1| + |BF_2| = 2a$
 即 $|AF_1| = 2a - 3m$, $|BF_1| = 2a - m$

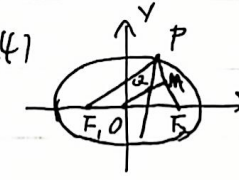
由 $\triangle AF_2O$ 中, $|AF_2| = a$, $|F_2O| = c$, 则 $\cos \angle AF_2O = \frac{c}{a}$ 由 $\angle AF_2F_1 + \angle BF_2F_1 = \pi$
 即 $\cos \angle AF_2F_1 = -\cos \angle BF_2F_1$ 则 $\frac{|F_1F_2|^2 + |AF_2|^2 - |AF_1|^2}{2|F_1F_2| \cdot |AF_2|} = -\frac{|F_1F_2|^2 + |BF_2|^2 - |BF_1|^2}{2|F_1F_2| \cdot |BF_2|}$
 则 $\frac{(2c)^2 + (3m)^2 - (2a-3m)^2}{2 \cdot 2c \cdot (3m)} = -\frac{(2c)^2 + m^2 - (2a-m)^2}{2 \cdot 2c \cdot m}$ 即 $m = \frac{2b^2}{3a}$
 由 $\cos \angle AF_2F_1 = \frac{(2c)^2 + (3m)^2 - (2a-3m)^2}{2 \cdot 2c \cdot 3m} = \frac{4c^2 - 4a^2 + 12mq}{12mc} = \frac{-4b^2 + 12mq}{12mc} = \frac{a}{2c}$ 又由 $\cos \angle AF_2F_1 = \frac{c}{a}$
 即 $\frac{a}{2c} = \frac{c}{a}$ 则 $a^2 = 2c^2$ 即 $e = \sqrt{\frac{c^2}{a^2}} = \sqrt{\frac{1}{2}} = \frac{\sqrt{2}}{2}$

(2)  由直线 l 斜率为 1, 则 $\angle AF_1F_2 = 45^\circ$, 设 $|BF_1| = m (m > 0)$
 由 $\overrightarrow{AF_1} = 3\overrightarrow{F_1B}$ 即 $|AF_1| = 3m$ 由第一定义, 则 $|AF_2| = 2a - 3m$,
 $|BF_2| = 2a - m$ 由 $\angle AF_1F_2 + \angle BF_1F_2 = \pi$
 即 $\frac{|AF_1|^2 + |F_1F_2|^2 - |AF_2|^2}{2|F_1F_2| \cdot |AF_1|} = \frac{|BF_1|^2 + |F_1F_2|^2 - |BF_2|^2}{2|BF_1| \cdot |F_1F_2|}$ 则 $\frac{(3m)^2 + (2c)^2 - (2a-3m)^2}{2 \times 3m \times 2c} = -\frac{m^2 + (2c)^2 - (2a-m)^2}{2 \times m \times 2c}$
 即 $m = \frac{2b^2}{3a}$ 由 $\angle AF_1F_2 = \frac{(2c)^2 + (3m)^2 - (2a-3m)^2}{2 \times 3m \times 2c} = \frac{a}{2c}$ 又由 $\angle AF_1F_2 = 45^\circ$ 即 $\frac{a}{2c} = \frac{\sqrt{2}}{2}$
 即 $e = \frac{\sqrt{2}}{2}$

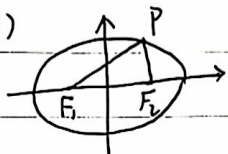
(3)  由椭圆的对称性, 不妨设点在第一象限 (第四象限也可)
 设过 A, F2 作 PF1 的垂足分别为 M, N. 由 $\triangle MF_1A \sim \triangle NF_1F_2$
 且由 A 是 OF_2 中点, 则 $|AM| = \frac{c}{2}$, 则 $|AF_1| = \frac{3}{2}c$, 即 $\frac{|AF_1|}{|AF_2|} = \frac{\frac{3}{2}c}{2c} = \frac{3}{4}$

则 $\frac{|AM|}{|NF_1|} = \frac{3}{4}$ 又由 $|AM| = \frac{1}{2}b$, 即 $|NF_1| = \frac{2}{3}b$ 由 $F_2 \perp X$ 轴, 则 P 点横坐标为 c.
 则 $\frac{c^2}{a^2} + \frac{y_p^2}{b^2} = 1$, 即 $y_p = \frac{b^2}{a}$ 则 $|PF_2| = \frac{b^2}{a}$ 由第一定义, 则 $|PF_1| = 2a - \frac{b^2}{a}$
 由等面积法, 则 $\frac{1}{2}|F_1F_2| \cdot |PF_2| = \frac{1}{2}|NF_1| \cdot |PF_1|$ 即 $\frac{1}{2} \cdot 2c \cdot \frac{b^2}{a} = \frac{1}{2} \cdot \frac{2}{3}b \cdot (2a - \frac{b^2}{a})$
 即 $3cb = 2a^2 - b^2$ 又由 $a^2 = b^2 + c^2$ 即 $b^2 + 2c^2 - 3bc = 0$ 即 $b = 2c$ 或 $b = c$.

当 $b = c$ 时, 由 $a^2 = b^2 + c^2$ 即 $a^2 = 2c^2$ 即 $e = \frac{\sqrt{2}}{2}$
 当 $b = 2c$ 时, 由 $a^2 = b^2 + c^2$ 即 $a^2 = 5c^2$ 即 $e = \frac{\sqrt{5}}{5}$ 综上 $e = \frac{\sqrt{2}}{2}$ 或 $\frac{\sqrt{5}}{5}$

(4)  设 OQ 与 PF2 交于点 M. 由 $OQ \parallel PF_1$ 且 $PF_1 \perp PF_2$ 则 $\angle QMP = 90^\circ$
 又由 PQ 是 $\angle F_1PF_2$ 的角平分线, 则 $\angle QPM = 45^\circ$ 即 $\triangle QMP$ 是等腰直角三角形, 则 $|QM| = |PM|$. 由 O 是 F_1F_2 中点, $OM \parallel PF_1$
 则 M 必为 PF_2 中点, 则 $|PM| = |MF_2|$ 设 $|PM| = m (m > 0)$ 则 $|QM| = |PM| = m$
 由 $|PF_2| = 2m$, 且由第一定义: $|PF_1| = 2a - 2m$ 由 $PF_1 \perp PF_2$ 则 $|PF_1|^2 + |PF_2|^2 = |F_1F_2|^2$
 则 $(2m)^2 + (2a - 2m)^2 = (2c)^2$ 又由 OM 是 PF_1 中位线, 则 $|QM| + |OM| = \frac{1}{2}|PF_1|$
 则 $b + m = (2a - 2m) \times \frac{1}{2}$ 即 $m = \frac{a-b}{2}$ 又由 $(2m)^2 + (2a - 2m)^2 = (2c)^2$
 即 $(a-b)^2 + (a+b)^2 = (2c)^2$ 则 $2a^2 = 3c^2$ 即 $e = \sqrt{\frac{c^2}{a^2}} = \sqrt{\frac{2}{3}} = \frac{\sqrt{6}}{3}$

(10)



设 $|PF_2| = m (m > 0)$, 则 $|PF_1| = \lambda m$. (其中 $\lambda \in [\frac{1}{3}, 3]$).

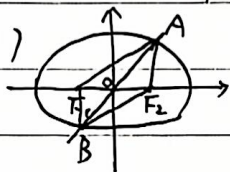
由第一定义: $m + \lambda m = 2a$ ① 由勾股定理: $m^2 + (\lambda m)^2 = (2c)^2$ ②

由 ②, 则 $\frac{(2c)^2}{(2a)^2} = \frac{m^2 + (\lambda m)^2}{(m + \lambda m)^2} = \frac{(\lambda^2 + 1)m^2}{(\lambda + 1)^2 m^2} = \frac{\lambda^2 + 1}{(\lambda + 1)^2}$ 设 $\lambda + 1 = t (t \in [\frac{4}{3}, 4])$

$$\text{则 } e^2 = \frac{(t-1)^2 + 1}{t^2} = \frac{t^2 - 2t + 2}{t^2} = 1 - \frac{2}{t} + \frac{2}{t^2} = 2\left[\frac{1}{t^2} - \frac{1}{t} + \left(\frac{1}{2}\right)^2\right] + 1 - 2 \times \left(\frac{1}{2}\right)^2 = 2\left(\frac{1}{t} - \frac{1}{2}\right)^2 + \frac{1}{2}$$

由 $\frac{1}{t} - \frac{1}{2} \in [-\frac{1}{4}, \frac{1}{4}]$, 则 $e^2 \in [\frac{1}{2}, \frac{5}{2}]$ 则 $e \in [\frac{\sqrt{2}}{2}, \frac{\sqrt{10}}{2}]$

(11)



由 $\vec{F_2A} \cdot \vec{F_2B} = (-\vec{AF_2}) \cdot (\vec{AF_2}) \leq \frac{4}{9}a^2$ 即 $\vec{AF_1} \cdot \vec{AF_2} \geq -\frac{4}{9}a^2$

由 $\vec{AF_1} \cdot \vec{AF_2} = \left(\frac{\vec{AF_1} + \vec{AF_2}}{2}\right) \cdot \left(\frac{\vec{AF_1} - \vec{AF_2}}{2}\right) = \frac{OA^2 - OF_2^2}{2} = \frac{OA^2 - c^2}{2}$

又由 $\angle F_1OA + \angle F_2OA = \pi$ 则 $\frac{AO^2 + OF_1^2 - AF_1^2}{2AO \cdot OF_1} = -\frac{AO^2 + OF_2^2 - AF_2^2}{2AO \cdot OF_2}$

设 $|F_2B| = m (m > 0)$, 则 $|AF_1| = m$, $|AF_2| = 2m$. 即 $AO^2 = \frac{5m^2 - 2c^2}{2} = \frac{5}{2}m^2 - c^2$

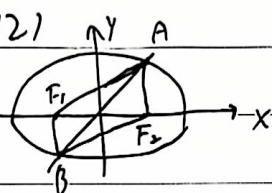
又由第一定义: $|AF_1| + |AF_2| = 2a$, 即 $2a = m + 2m$, 则 $m = \frac{2}{3}a$.

则 $AO^2 = \frac{5}{2}m^2 - c^2 = \frac{10}{9}a^2 - c^2$ 则 $OA^2 - c^2 \geq -\frac{4}{9}a^2$ 则 $OA^2 \geq -\frac{4}{9}a^2 + c^2$

则 $\frac{10}{9}a^2 - c^2 \geq -\frac{4}{9}a^2 + c^2$ 则 $\frac{16}{9}a^2 \geq 2c^2$ 则 $\frac{8}{9} \geq e^2$ 又由 $e \in (0, 1)$

则 $e \in (0, \frac{2\sqrt{2}}{3}]$

(12)



设 $|BF_1| = m$, $|AF_1| = n (m, n > 0)$, 又由 $|BF_1| = |AF_2|$, 则 $|AF_2| = m$

又由第一定义: $|AF_1| + |AF_2| = 2a$, 则 $m + n = 2a$.

又由 $AF_1 \perp BF_1$, 则 $AF_1 \perp AF_2$ 则 $m^2 + n^2 = (2c)^2$

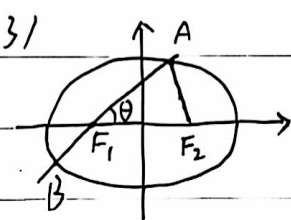
则 $e^2 = \frac{(2c)^2}{(2a)^2} = \frac{m^2 + n^2}{(m + n)^2} = \frac{m^2 + n^2}{m^2 + n^2 + 2mn}$ 设 $\frac{m}{n} = \lambda (\lambda > 0)$

$$\text{则 } e^2 = \frac{(\frac{m}{n})^2 + 1}{(\frac{m}{n})^2 + 2\frac{m}{n} + 1} = \frac{\lambda^2 + 1}{\lambda^2 + 2\lambda + 1} = \frac{\lambda^2 + 1}{(\lambda + 1)^2}$$

$$\text{设 } \lambda + 1 = t (t > 1) \text{ 则 } e^2 = \frac{(t-1)^2 + 1}{t^2} = \frac{t^2 - 2t + 2}{t^2} = 1 - \frac{2}{t} + \frac{2}{t^2} = 2\left[\frac{1}{t^2} - \frac{1}{t} + \left(\frac{1}{2}\right)^2\right] + 1 - 2 \times \left(\frac{1}{2}\right)^2 = 2\left(\frac{1}{t} - \frac{1}{2}\right)^2 + \frac{1}{2}$$

由 $t > 1$, $\frac{1}{t} < 1$, 则 $\frac{1}{t} - \frac{1}{2} < \frac{1}{2}$ 所以 $e^2 \in (\frac{1}{2}, 1)$ 即 $e \in (\frac{\sqrt{2}}{2}, 1)$

(13)



由 $|AF_1| = |F_2B| = 2c$, 由第一定义: $|AF_2| = 2a - 2c$

由 $\cos \theta = \frac{|AF_1|^2 + |F_2B|^2 - |AF_2|^2}{2|AF_1| \cdot |F_2B|} = \frac{(2c)^2 + (2c)^2 - (2a - 2c)^2}{2 \cdot 2c \cdot 2c} = \frac{4c^2 + 4c^2 - 4a^2 + 8ac - 4c^2}{8c^2} = \frac{4c^2 + 8ac - 4a^2}{8c^2}$

又由 $\theta \in [\frac{\pi}{6}, \frac{\pi}{3}]$ 则 $\cos \theta \in [\frac{1}{2}, \frac{\sqrt{3}}{2}]$

则 $\frac{1}{2} \leq \frac{e^2 + 2e - 1}{2e^2} \leq \frac{\sqrt{3}}{2}$ 则 $\begin{cases} \frac{1}{2} \leq \frac{e^2 + 2e - 1}{2e^2} \\ \frac{e^2 + 2e - 1}{2e^2} \leq \frac{\sqrt{3}}{2} \end{cases}$ 则 $\begin{cases} 0 \leq 2e - 1 \\ (\sqrt{3} - 1)e^2 - 2e + 1 \geq 0 \end{cases}$

由 $(\sqrt{3} - 1)e^2 - 2e + 1 \geq 0$

则该方程零点

$$e = \frac{2 \pm \sqrt{(-2)^2 - 4(\sqrt{3} - 1)}}{2(\sqrt{3} - 1)} = \frac{2 \pm \sqrt{8 - 4\sqrt{3}}}{2(\sqrt{3} - 1)}$$

$$= \frac{2 \pm \sqrt{4 - 2\sqrt{3}}}{2(\sqrt{3} - 1)} = \frac{2 \pm \sqrt{2(\sqrt{3} - 1)}}{2(\sqrt{3} - 1)} = \frac{2 \pm \sqrt{2}(\sqrt{3} - 1)}{2(\sqrt{3} - 1)} = \frac{1}{\sqrt{3} - 1} \pm \frac{\sqrt{2}}{2} = \frac{\sqrt{3} + 1}{2} \pm \frac{\sqrt{2}}{2}$$

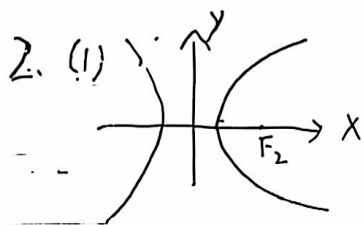
由 $e \in (0, 1)$

$$\text{则 } e = \frac{\sqrt{3} + 1 - \sqrt{2}}{2}$$

... 则 $\begin{cases} e \geq \frac{\sqrt{3} + 1 - \sqrt{2}}{2} \\ e \geq \frac{1}{2} \\ e \in (0, 1) \end{cases}$

则 $e \in [\frac{\sqrt{3} + 1 - \sqrt{2}}{2}, 1)$

IV

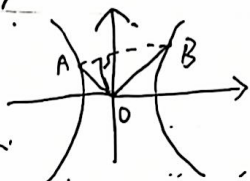


由渐近线方程: $\frac{x}{a} \pm \frac{y}{b} = 0$, 不妨取 $y = \frac{b}{a}x$, 由 $F_2(c, 0)$

由 $d = \frac{|\frac{b}{a}c|}{\sqrt{1+\frac{b^2}{a^2}}} = \frac{|bc|}{\sqrt{a^2+b^2}} = \frac{|bc|}{|c|} = |b| = b$ 又由 $d = a$

则 $a = b$ 则 $a^2 = b^2$, 又由 $c^2 = a^2 + b^2 = 2a^2$ 即 $e = \sqrt{2}$

(2)



设直线 $AB = y = kx + m$ 联立 $\begin{cases} y = kx + m \\ \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \end{cases}$ 则

$(b^2 - a^2k^2)x^2 - 2kma^2x - a^2(m^2 + b^2) = 0$ 由题 $b^2 - a^2k^2 \neq 0$

则 $x_1 + x_2 = \frac{2kma^2}{b^2 - a^2k^2}$, $x_1x_2 = \frac{-a^2(m^2 + b^2)}{b^2 - a^2k^2}$

由 $y_1y_2 = (kx_1 + m)(kx_2 + m) = k^2x_1x_2 + km(x_1 + x_2) + m^2 =$
 $\frac{-a^2k^2(m^2 + b^2) + km(2kma^2) + m^2(b^2 - a^2k^2)}{b^2 - a^2k^2} = \frac{b^2(m^2 - a^2k^2)}{b^2 - a^2k^2}$

由 $OA \perp OB$, 则 $\vec{OA} \cdot \vec{OB} = 0$. 则 $x_1x_2 + y_1y_2 = 0$. 则 $-a^2(m^2 + b^2) + b^2(m^2 - a^2k^2) = 0$

则 $m^2(b^2 - a^2) = a^2b^2(k^2 + 1)$ 又由 $d_{O-AB} = \frac{|m|}{\sqrt{1+k^2}}$ 且 $d_{O-AB} = c$

则 $m^2 = (1+k^2)c^2$ 则 $\frac{m^2}{k^2+1} = \frac{a^2b^2}{b^2-a^2} = c^2$ 则 $\frac{a^2(c^2-a^2)}{c^2-a^2-a^2} = c^2$

则 $a^2(c^2 - a^2) = c^2(c^2 - 2a^2)$ 则 $c^4 - 3a^2c^2 + a^4 = 0$ 则 $e^4 - 3e^2 + 1 = 0$

则 $e^2 = \frac{3 \pm \sqrt{3^2 - 4 \times 1}}{2} = \frac{3 \pm \sqrt{5}}{2}$ 又由 $e \in (1, +\infty)$ 则 $e^2 = \frac{3 + \sqrt{5}}{2} = \frac{6 + 2\sqrt{5}}{4}$

则 $e = \frac{\sqrt{6+2\sqrt{5}}}{2} = \frac{\sqrt{5}+1}{2}$

(3) 由渐近线: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$ 由 $\begin{cases} \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0 \\ x - 3y + m = 0 \end{cases}$ 则 $(9b^2 - a^2)y^2 - 6mb^2y + m^2b^2 = 0$

$y_1 + y_2 = \frac{6mb^2}{9b^2 - a^2}$, $y_1y_2 = \frac{m^2b^2}{9b^2 - a^2}$ 易知 P 点必在 AB 的中垂线上.

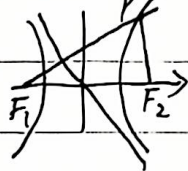
又 $x_1 + x_2 = 3y_1 - m + 3y_2 - m = 3(y_1 + y_2) - 2m = \frac{2ma^2}{9b^2 - a^2}$ 则设 AB 中点为 C.

则 $PC \perp AB$. 由 $C(\frac{ma^2}{9b^2 - a^2}, \frac{3mb^2}{9b^2 - a^2})$, 易知 $K_{AB} = \frac{1}{3}$, 由 $K_{PC} \cdot K_{AB} = -1$

则 $K_{PC} = -3$. 则 $K_{PC} = \frac{\frac{3mb^2}{9b^2 - a^2} - 0}{\frac{ma^2}{9b^2 - a^2} - m} = \frac{3mb^2}{2ma^2 - 9mb^2} = \frac{3b^2}{2a^2 - 9b^2}$ 则 $\frac{3b^2}{2a^2 - 9b^2} = -3$

则 $a^2 = 4b^2 = 4(c^2 - a^2)$ 则 $5a^2 = 4c^2$ 即 $e = \frac{\sqrt{5}}{2}$

(4) 由双曲线的对称性 不妨取渐近线: $\frac{x}{a} + \frac{y}{b} = 0$ 的垂线.



由渐近线斜率为 $K_1 = -\frac{b}{a}$ 则直线 PF_1 斜率 $K_2 = \frac{a}{b}$

由正弦定理: $\frac{|PF_1| - |PF_2|}{\sin \angle F_1F_2P} = \frac{|F_1F_2|}{\sin \angle F_1PF_2}$ 又由 $\angle F_1PF_2 = 45^\circ$,

且 $\sin \angle PF_1F_2 = \frac{a}{c}$, $\cos \angle PF_1F_2 = \frac{b}{c}$, $\angle F_1F_2P = \pi - \frac{\pi}{4} - \angle PF_1F_2 = \frac{3}{4}\pi - \angle PF_1F_2$

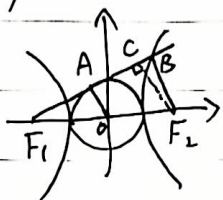
由第一定义: $|PF_1| - |PF_2| = 2a$ $\frac{2a}{2c} = \frac{1}{e} = \frac{\sin \angle F_1F_2P - \sin \angle PF_1F_2}{\sin \frac{\pi}{4}}$

设 $\angle PF_1F_2 = \theta$. 则 $\sin \theta \sin \frac{\pi}{4} = \sin(\frac{3}{4}\pi - \theta) + \sin \theta = \sin \frac{3}{4}\pi \cos \theta - \cos \frac{3}{4}\pi \sin \theta + \sin \theta$

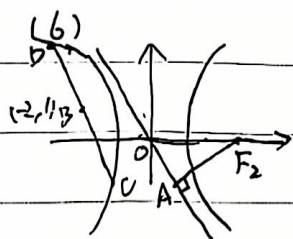
则 $\frac{\sqrt{2}}{2} \cos \theta = \sin \theta$, 则 $\tan \theta = \frac{\sqrt{2}}{2} = \frac{a}{b}$ 则 $\frac{a^2}{b^2} = \frac{1}{2}$ 则 $2a^2 = b^2 = c^2 - a^2$

则 $3a^2 = c^2$ 则 $e^2 = 3$, 则 $e = \sqrt{3}$.

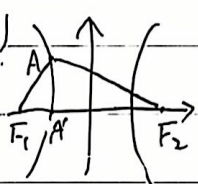
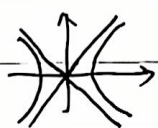
(5)

过 F_2 作直线 l 垂线, 交 l 于点 C . 由 $\triangle AF_1O \sim \triangle CF_1F_2$ 且 $\frac{|OF_1|}{|F_1F_2|} = \frac{c}{2c} = \frac{1}{2} = \frac{|OA|}{|CF_2|}$ 又由 $|OA| = a$ 则 $|CF_2| = 2a$.又由第一定义: $|BF_1| - |BF_2| = 2a$. 由 $|CF_1| = \sqrt{|F_1F_2|^2 - |CF_2|^2} = 2b$ 则 $|BC| = |BF_1| - |CF_1| = 2a + |BF_2| - 2b$ 由勾股定理: $|BC|^2 + |CF_2|^2 = |BF_2|^2$ 则 $(2a - 2b + |BF_2|)^2 + (2a)^2 = |BF_2|^2$ 则 $(2a - 2b)^2 + 2(2a - 2b)|BF_2| + (2a)^2 = 0$ 即 $|BF_2| = \frac{b^2 - 2ab + 2a^2}{b - a}$ 由 $|AB| = 2|AF_1|$, 则 $|BF_1| = |AB| + |AF_1| = 3|AF_1| = 3b$ 则 $3b - \frac{b^2 - 2ab + 2a^2}{b - a} = 2a$ 则 $(3b - 2a)(b - a) = b^2 - 2ab + 2a^2$ 则 $3b^2 - 5ab + 2a^2 = b^2 - 2ab + 2a^2$ 则 $2b = 3a$ 则 $4b^2 = 9a^2$ 又由 $b^2 = c^2 - a^2$ 则 $4c^2 = 13a^2$ 则 $e^2 = \frac{13}{4}$ 则 $e = \frac{\sqrt{13}}{2}$

(6)

由题, 易得, 直线 l 垂直于: 渐近线: $\frac{x}{a} + \frac{y}{b} = 0$ 由 $AF_2 \perp$ 渐近线, $AF_2 \perp CD$, 则 $CD \parallel$ 渐近线.则 $K_{CB} = -\frac{b}{a}$, 又由 $K_{CD} \cdot K_{OB} = e^2 - 1 = \frac{c^2}{a^2} - 1 = \frac{c^2 - a^2}{a^2} = \frac{b^2}{a^2}$ 又由 $K_{OB} = -\frac{1}{2}$ 则 $-\frac{b}{a} \cdot (-\frac{1}{2}) = \frac{b^2}{a^2}$ 则 $\frac{b}{2a} = \frac{b^2}{a^2}$ 则 $\frac{1}{2} = \frac{b}{a}$ 则 $a = 2b$ 则 $a^2 = 4b^2 = 4(c^2 - a^2)$ 则 $5a^2 = 4c^2$ 则 $e^2 = \frac{5}{4}$ 所以 $e = \frac{\sqrt{5}}{2}$

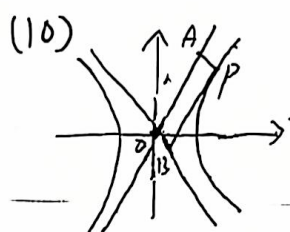
(7)

设 $\angle F_1AF_2 = \theta$, 由 $\cos \theta = \frac{|AF_1|^2 + |AF_2|^2 - |F_1F_2|^2}{2|AF_1| \cdot |AF_2|}$ 又由 $|AF_2| = 7|AF_1|$ 且设 $|AF_1| = m (m > 0)$, 则 $\cos \theta = \frac{m^2 + (7m)^2 - (2c)^2}{2m \cdot 7m} = \frac{50m^2 - 4c^2}{14m^2}$ 又由 $|AF_2| - |AF_1| = 2a$ 则 $6m = 2a$ 则 $m = \frac{1}{3}a$ 则 $\cos \theta = \frac{\frac{50}{9}a^2 - 4c^2}{14 \cdot \frac{1}{9}a^2} = \frac{50a^2 - 36c^2}{14a^2} = \frac{25}{7} - \frac{18}{7}e^2$ 又由 $\theta \in (0, \pi)$ 则 $\cos \theta \in [-1, 1)$ 则 $-1 \leq \frac{25}{7} - \frac{18}{7}e^2 < 1$ 则 $e \in (1, \frac{4}{3}]$ (8) 由 $y = 2x$ 与双曲线 E 有交点, 则渐近线的斜率必大于 2.由 $\frac{x}{a} - \frac{y}{b} = 0$ 的斜率为 $k = \frac{b}{a}$ 则 $\frac{b}{a} > 2$ 则 $b > 2a$ 则 $b^2 > 4a^2$ 又由 $b^2 = c^2 - a^2$ 则 $c^2 - a^2 > 4a^2$ 即 $c^2 > 5a^2$ 则 $e^2 > 5$ 则由 $e \in (1, +\infty)$, 则 $e > \sqrt{5}$

(9)

由直线 $l: y = \sqrt{3}(x - c)$, 联立 $\begin{cases} \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \\ y = \sqrt{3}(x - c) \end{cases}$ 则 $(b^2 - 3a^2)x^2 + 6a^2cx - 3a^2c^2 - a^2b^2 = 0$ 由与右支有两交点 则不妨设 $A(x_1, y_1), B(x_2, y_2)$ 则 $\begin{cases} x_1x_2 > 0 \\ x_1 + x_2 > 0 \\ b^2 - 3a^2 \neq 0 \end{cases}$ 则 $\begin{cases} -\frac{a^2(b^2 + 3c^2)}{b^2 - 3a^2} > 0 \\ \frac{-6a^2c}{b^2 - 3a^2} > 0 \\ b^2 - 3a^2 \neq 0 \end{cases}$ 则 $3a^2 - b^2 > 0$ 又由 $b^2 = c^2 - a^2$ 则 $3a^2 > c^2 - a^2$ 则 $4a^2 > c^2$ 则 $e^2 < 4$ 又由 $e > 1$ 则 $e \in (1, 2)$

VI



(10) 设 $P(x_0, y_0)$, 则 $\frac{x_0^2}{a^2} - \frac{y_0^2}{b^2} = 1$ 则 $b^2 x_0^2 - a^2 y_0^2 = a^2 b^2$

易知 OA、OB 直线斜率分别为: $\frac{b}{a}$ 和 $-\frac{b}{a}$

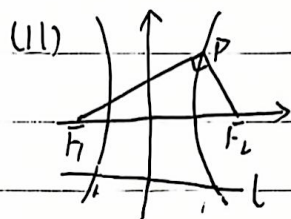
则 AP 直线: $y = \frac{b}{a}(x - x_0) + y_0$ 联立 $\begin{cases} y = \frac{b}{a}(x - x_0) + y_0 \\ y = -\frac{b}{a}x \end{cases}$ 和 $\begin{cases} y = -\frac{b}{a}(x - x_0) + y_0 \\ y = \frac{b}{a}x \end{cases}$

BP 直线: $y = \frac{b}{a}(x - x_0) + y_0$

则 $A(\frac{bx_0 - ay_0}{2b}, -\frac{bx_0 - ay_0}{2a})$, $B(\frac{bx_0 + ay_0}{2b}, \frac{bx_0 + ay_0}{2a})$

则 $\overrightarrow{OA} \cdot \overrightarrow{OB} = \frac{b^2 x_0^2 - a^2 y_0^2}{4b^2} - \frac{b^2 x_0^2 - a^2 y_0^2}{4a^2} = \frac{a^2 b^2}{4b^2} - \frac{a^2 b^2}{4a^2} = \frac{a^2 - b^2}{4}$ 则 $\frac{a^2 - b^2}{4} \geq \frac{b^2}{4}$

则 $a^2 \geq 2b^2 = 2(c^2 - a^2)$ 则 $3a^2 \geq 2c^2$ 则 $e \leq \frac{\sqrt{6}}{2}$ 又由 $e > 1$ 则 $e \in (1, \frac{\sqrt{6}}{2}]$



联立 $\begin{cases} 2x + 3y - 2c = 0 \\ \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \end{cases}$ 则 $(b^2 - \frac{4}{9}a^2)x^2 + \frac{8}{9}a^2cx - a^2(\frac{4}{9}c^2 + b^2) = 0$

由左、右支各有一交点, 则 $x_1 x_2 < 0$

则 $\frac{-a^2(\frac{4}{9}c^2 + b^2)}{b^2 - \frac{4}{9}a^2} < 0$ 则 $b^2 - \frac{4}{9}a^2 > 0$ 又由 $b^2 = c^2 - a^2$

则 $e^2 > \frac{13}{9}$ 则 $e > \frac{\sqrt{13}}{3}$ 设 $|PF_1| = m, |PF_2| = n (m, n > 0)$

由第一定义: $m - n = 2a$; 由勾股定理: $m^2 + n^2 = (2c)^2$, 又由 $m > 2n$ 则 $m - n > n$

则 $n < 2a$. 由 $m = 2a + n$, 则 $(n + 2a)^2 + n^2 = (2c)^2$ 则 $n^2 + 2an + a^2 = 2c^2 - a^2$

则 $(n + a)^2 = 2c^2 - a^2$ 又由 $n < 2a$, 则 $n + a < 3a$. 则 $2c^2 - a^2 < 9a^2$

则 $e^2 < 5$ 则 $e < \sqrt{5}$ 即 $e \in (\frac{\sqrt{13}}{3}, \sqrt{5})$

(12) 由圆 $C: (x+2)^2 + (y-4)^2 = 4$, 且 $|F_1 F_2| = 4$, 则 $c = 2$

则 $C(-2, 4)$ 设 $M(x_0, y_0)$ $y_0 \in [2, 6]$

若四边形 $OMNF_2$ 为平行四边形, 则 MF_2 的中点必在渐近线上.

如果 N 点在: 渐近线 $\frac{x}{a} - \frac{y}{b} = 0$ 上. 则记 MF_2 中点为 R.

则 $R(\frac{x_0 + 2}{2}, \frac{y_0}{2})$ 则 $\frac{x_0 + 2}{2a} = \frac{y_0}{2b}$ 则 $y_0 = \frac{b}{a}(x_0 + 2)$

又由 $(x_0 + 2)^2 + (y_0 - 4)^2 = 4$ 则 $(\frac{ax_0}{b})^2 + (y_0 - 4)^2 = 4$

则 $\frac{a^2}{b^2} y_0^2 = 4 - (y_0 - 4)^2$ 则 $\frac{a^2}{b^2} = \frac{4 - (y_0 - 4)^2}{y_0^2}$ 则 $\frac{b^2}{a^2} = \frac{y_0^2}{4 - (y_0 - 4)^2}$ 又由 $e^2 - 1 = \frac{b^2}{a^2}$

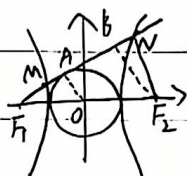
则 $e^2 - 1 = \frac{y_0^2}{y_0^2 + 8y_0 - 12} = \frac{1}{\frac{y_0^2}{y_0^2 + 8y_0 - 12} - 1}$ 设 $t = \frac{1}{y_0} (t \in [\frac{1}{6}, \frac{1}{2}])$

则 $e^2 - 1 = \frac{1}{-12(t - \frac{1}{3})^2 + \frac{1}{3}} \in [3, +\infty)$ 则 $e \in [2, +\infty)$

同理, 若 N 在渐近线: $\frac{x}{a} + \frac{y}{b} = 0$ 上, 则 $\frac{b}{a} = \frac{-y_0}{x_0 + 2}$ 则 $e^2 - 1 = (\frac{-y_0}{x_0 + 2})^2$
 $= \frac{y_0^2}{(x_0 + 2)^2} = \frac{y_0^2}{4 - (y_0 - 4)^2}$ 同理, 得 $e \in [2, +\infty)$.

综上 $e \in [2, +\infty)$.

(13) 不妨设 N 在第一象限, 且切线与圆 O 切点为 A , 过 F_2 作切线的垂线, 垂足为 B . 由 $|OA| = a$, 则 $|BF_2| = 2a$.



由第一定义: $|NF_1| - |NF_2| = 2a$. 又由 $|BF_2| = \sqrt{|F_2N|^2 - |NF_2|^2} = 2b$

则 $|NB| = |NF_1| - |BF_2| = 2a + |NF_2| - 2b$ 又由 $\cos \angle F_1NF_2 = \frac{3}{5}$

则 $\frac{|NB|}{|NF_2|} = \frac{2a - 2b + |NF_2|}{|NF_2|} = \frac{3}{5}$, 则 $\frac{2}{5}|NF_2| = 2b - 2a$ 则 $|NF_2| = 5(b - a)$

则 $|NF_1| = 2a + |NF_2| = 5b - 3a$ 由 $\cos \angle F_1NF_2 = \frac{|NF_1|^2 + |NF_2|^2 - |F_1F_2|^2}{2|NF_1| \cdot |NF_2|} = \frac{(5b-3a)^2 + (5b-5a)^2 - (2c)^2}{2(5b-3a)(5b-5a)} = \frac{3}{5}$

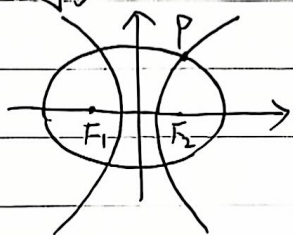
则 $3a^2 - 8ab + 4b^2 = 0$ 则 $(a-2b)(3a-2b) = 0$ 即 $a = 2b$ 或 $3a = 2b$

当 $a = 2b$ 时, 则 $a^2 = 4b^2 = 4(c^2 - a^2)$, 则 $4c^2 = 5a^2$ 则 $e^2 = \frac{5}{4}$, 则 $e = \frac{\sqrt{5}}{2}$

当 $a = \frac{2}{3}b$ 时, 则 $a^2 = \frac{4}{9}b^2 = \frac{4}{9}(c^2 - a^2)$, 则 $\frac{4}{9}c^2 = \frac{13}{9}a^2$ 则 $e^2 = \frac{13}{4}$, 则 $e = \frac{\sqrt{13}}{2}$

综上 $e = \frac{\sqrt{5}}{2}$ 或 $\frac{\sqrt{13}}{2}$

例 3:



(1) 设 $|PF_1| = m (m > 0)$, $|PF_2| = n (n > 0)$.

则由第一定义: $\begin{cases} |PF_1| + |PF_2| = 2a_1 \\ |PF_1| - |PF_2| = 2a_2 \end{cases}$ 则 $\begin{cases} m + n = 2a_1 \\ m - n = 2a_2 \end{cases}$

则 $\begin{cases} m = a_1 + a_2 \\ n = a_1 - a_2 \end{cases}$ 由 $PF_2 \perp x$ 轴, 则 $m^2 = n^2 + (2c)^2$

则 $m^2 - n^2 = (m+n)(m-n) = 4a_1a_2 = 4c^2$ 则 $\frac{c^2}{a_1a_2} = 1$

则 $e_1e_2 = 1$

(2) 由 $\angle F_1PF_2 = 60^\circ$, 则 $\cos \angle F_1PF_2 = \frac{m^2 + n^2 - (2c)^2}{2mn} = \frac{(a_1+a_2)^2 + (a_1-a_2)^2 - (2c)^2}{2(a_1+a_2)(a_1-a_2)} = \frac{2a_1^2 + 2a_2^2 - 4c^2}{2a_1^2 - 2a_2^2} = \frac{1}{2}$ 则 $a_1^2 + 3a_2^2 = 4c^2$ 则 $\frac{a_1^2}{c^2} + \frac{3a_2^2}{c^2} = 4$

由 $e_1^2 + e_2^2 = \frac{c^2}{a_1^2} + \frac{c^2}{a_2^2} = (\frac{c^2}{a_1^2} + \frac{c^2}{a_2^2}) (\frac{a_1^2}{c^2} + \frac{3a_2^2}{c^2}) \times \frac{1}{4} = \frac{1}{4} (1 + 3 + \frac{a_1^2}{a_2^2} + \frac{3a_2^2}{a_1^2}) \geq \frac{1}{4} (4 + 2\sqrt{\frac{a_1^2}{a_2^2} \cdot \frac{3a_2^2}{a_1^2}})$

$= 1 + \frac{\sqrt{3}}{2}$ (当且仅当 $\frac{a_1^2}{a_2^2} = \frac{3a_2^2}{a_1^2}$ 时取等, 即 $a_1^2 = \sqrt{3}a_2^2$ 时取等)

(3) 由 $|PF_2| = |F_1F_2|$, 则 $a_1 - a_2 = 2c$, 则 $\frac{1}{e_1} - \frac{1}{e_2} = 2$ 则 $\frac{1}{e_1} = 2 + \frac{1}{e_2} = \frac{2e_2 + 1}{e_2}$

则 $e_1 = \frac{e_2}{2e_2 + 1}$ 则 $e_1e_2 = \frac{e_2^2}{2e_2 + 1} = \frac{1}{\frac{1}{e_2^2} + \frac{2}{e_2}} = \frac{1}{(\frac{1}{e_2} + 1)^2 - 1}$ 又由 $e_2 > 1$ 则 $0 < \frac{1}{e_2} < 1$

则 $1 < \frac{1}{e_2} + 1 < 2$ 则 $(\frac{1}{e_2} + 1)^2 - 1 \in (0, 3)$ 则 $e_1e_2 \in (\frac{1}{3}, +\infty)$

VIII

例4:

(1)

$$\text{由 } r_{\text{内}} = \frac{2S_{\triangle PF_1F_2}}{C_{\triangle PF_1F_2}} = \frac{2c \cdot y_1}{2a+2c} \quad \text{又由 } r_{\text{内}} = y_2$$

$$\text{则 } y_2 = \frac{2c y_1}{2a+2c} = \frac{c y_1}{a+c} \quad \text{又由 } y_1 = 3y_2$$

$$\text{则 } \frac{y_1}{y_2} = \frac{a+c}{c} = \frac{1}{e} + 1 = 3 \quad \text{即 } e = \frac{1}{2}$$

(2) 由切线长定理: $|PA| = |PB|$, $|AF_1| = |CF_1|$, $|BF_2| = |CF_2|$

又由 $|PF_1| + |PF_2| = 2a$. 则 $|PF_1| + |PF_2| = (|PA| + |AF_1|) + (|PB| +$

$$|BF_2|) = (|PA| + |CF_1|) + (|PB| + |CF_2|) = |PA| + |PB| + |CF_1| + |CF_2|$$

又由 $|CF_1| + |CF_2| = 2c$, 则 $|PF_1| + |PF_2| = |PA| + |PB| + 2c = 2a$ 即 $|PA| + |PB| = 2a - 2c$

又由 $\triangle PF_1F_2$ 的周长为 $2a + 2c$, 则 $\frac{2a+2c}{2a-2c} = 3$. 即 $\frac{a+c}{a-c} = \frac{1+e}{1-e} = 3$ 则 $e = \frac{1}{2}$

(3) 由 $|CF_1| + |CF_2| = 2c$ 且 $|CF_1| = x_2 - (-c) = x_2 + c$, $|CF_2| = x_2 - c$.

又由 $|PA| = a - c$, 且 $|AF_1| = |CF_1|$, 即 $|PF_1| = |PA| + |AF_1| = a - c + x_2 + c = a + x_2$

又由 $|PF_1| = a + ex_1$ 则 $|PF_1| = a + ex_2$ 即 $2ex_2 = x_2$ 即 $e = \frac{x_2}{2x_2} = \frac{1}{2}$

